

Scenario D: Long-Job Sensitivity Case

Input

Quantum = 2

Process	Arrival Time	Burst Time
P1	0	20
P2	1	2
P3	2	1
P4	3	2

Analysis

- **Lower average waiting time:** Round Robin.
- **Lower response time:** Round Robin
- **Fairness:** Round Robin is clearly fair (equal CPU sharing)
- **Short job efficiency:** In this case SJF did not complete short jobs efficiently because the long process started first and blocked all shorter processes until it finished.
- **Quantum effect:** Allowed processes to share CPU time frequently
- **Recommendation:**
Round Robin is recommended for this workload because it prevents long processes from blocking shorter ones and provides better fairness and responsiveness.

Conclusion

- Round Robin performed better in both average waiting time and average response time.
- Round Robin appeared more balanced because all processes were given CPU time in a fair and cyclic manner.
- SJF was not efficient in this scenario because it allowed the long process to execute first, causing high delays for all other processes.
- The selected quantum improved system responsiveness by allowing frequent context switching. It ensured that no process waited too long and reduced the impact of long processes on the system.

CPU Scheduling Simulator

CPU Scheduling Simulator

InputGantt ChartsMetricsComparison

Process Input Configuration

No. of Processes:4

Time Quantum:2

Generate Table

Process ID	Arrival Time	Burst Time
P1	0	20
P2	1	2
P3	2	1
P4	3	2

Clear Data

Run Simulation



